Mercedes-Benz

Passenger Car Production Planning
Assembly

Production Planning
Assembly

Standard specification assembly
non-project-specific - part 1

Version

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10	4.4. Technical Safety Approval and	37	New milestone	3rd interim payment for milestone "Process approval of operational system" postponed	60.02	2017.11.14	Zehm/Frede
11	4.6. Process approval of operational system	38	New milestone	"Process approval of operational system" added as new milestone.	60.02	2017.11.14	Zehm/Frede
12	2.6.2. Green Production	13	Green Production	Addendum to chapter re. energy efficiency	60.03	26.01.2018	Zehm/Frede
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Foreword

The total tender consists of 3 volumes.

- Volume 1 - Standard invitation to bid (non-project-specific part)
- Volume 2 - Project-specific part
- Volume 3 - Project specifications
- Mercedes-Benz AG Group specifications (e.g. DBL 9606)

By submitting a tender, the supplier acknowledges the terms & conditions in Volumes 1, 2 and 3 and undertakes to download these once ECM access has been set up (see Volume 2), and to comply with them when preparing the tender and after any subsequent contract award.

When making awards, Mercedes-Benz AG reserves the right only to evaluate and consider tenders that have been formulated and signed in legally binding form on the forms prescribed by Mercedes-Benz AG in Volume 2.

The requirements formulated in this non project-specific part of the standard tender do not absolve the supplier of his responsibility for preparing the annexes and documents to be supplied by him in a professional manner, and in compliance with the laws, regulations and requirements in force at the production and installation location in question.

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1. Document delivery, submitted samples and nondisclosure

The supplier hereby confirms receipt of the planning documentation for the above project in paper or digital form. Mercedes-Benz AG has a decisive commercial interest in preventing third parties from obtaining any knowledge whatsoever of the appearance or technical details of the sample parts made available.

These documents and samples must therefore be stored at a safe location inaccessible to unauthorized persons. They must not be made accessible to third parties or used for commercial purposes without the express written consent of Mercedes-Benz AG.

If the bidder is provided with parts of the planning documents in the original file format, e.g. in order to assist him with translation into his national language, such files must not be passed on to his employees or sub-suppliers as original files.

He undertakes not to change the content or wording of all project-related information, provided samples, component data, documentation, recorded conversations and other materials, to use them only for this project and to treat them as strictly confidential. The bidder will confirm a confidentiality agreement by appending his signature.

If no contract is concluded on the basis of this bid, all information, including all copies, shall be destroyed in their entirety within one month of notification of the decision by Mercedes-Benz AG Procurement.

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2. General Project Conditions

The drawing-up of the bid including any alternatives and the bidder's expenses for visits, developments, project documents, layouts, etc. shall be free of charge for Mercedes-Benz AG.

2.1. Basic Principles

The following are basic principles of this tender and must be taken into account in the bid:

- all non project-specific and project-specific provisions described or referenced in Volumes 1 to 3 of this tender,
- any additional requirements,
- all component data, drawings, sketches and layout documentation,
- the general procurement terms and conditions of Mercedes-Benz AG.

These principles are referred to collectively in the following contractual provisions. The contents of these contractual represent an overall tender for contract award to a supplier. It is therefore not permissible to consider individual parts of the specification separately; instead the entire context shall always be included.

The overall project price is a fixed price and is composed of the prices of the individual items.
As part of the decision-making process when awarding the contract, and at the discretion of Mercedes-Benz AG, individual items of the offered scope of delivery may be completely or partially omitted without any entitlement to compensation or replacement on the part of the bidder.

These principles also apply to all modifications or additional, project-related products/services requested of the bidder after contract award.

2.2. Project Specifications, Laws, Standards, Regulations

The bidder is responsible for ensuring that his performance under the contract, and in particular the provided equipment, systems, machinery and activities comply with the accepted technical rules and all applicable laws, standards, regulations, requirements, etc. at the time of acceptance.
All national regulations of the country in which the services are provided shall be taken into account.

In this context, it is expressly pointed out that the contractual provisions have contents that require explicit, detailed training for the described situations. The expense incurred for this is part of the order and is included in the quoted price.

With regard to content, the contractual provisions are divided into documents that must be taken into account for the delivery of new systems and those that play a role in the conversion of already existing installations. For installations to be modified, the responsible department of the installation location shall be contacted about the system-specific interpretation of the project requirements.

It is always necessary to fulfill the superordinate legal requirements and standards, e.g. EU Machinery Directive, that apply at the installation location of production equipment.

Moreover, the specific standards and specifications of Mercedes-Benz AG shall be complied with. The bidder is obliged to obtain the latest version of these standards and specifications from Mercedes-Benz AG if he is not already familiar with them. In all cases, the bidder shall always ensure that the standards and specifications he has are applicable and up-to-date.

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To keep the number of spare and wearing parts to a minimum, only components listed in the approved materials lists (AML) may be used. If these standard components are used on a large scale and if several standards are approved in the AFL, coordination meetings shall be conducted after the order placement to minimize the standard components. If there are several suppliers, they are obliged to agree upon a standard under the guidance of Mercedes-Benz AG. This applies, for example, to protective devices, pneumatic clamps, etc.

If there are areas with a risk of explosion, the corresponding regulations shall be observed, e.g. ATEX Guideline.

Changes to the legal situation which occur after contract placement shall always be taken into account. Additional expense claims by the bidder are to be submitted in the form of a supplementary bid. Changes to contractual provisions, laws, standards and regulations following submission of the bid shall be examined by the bidder for their effects on costs if the bid acceptance deadline has passed. The documents to be taken into account are divided into the following areas:

- Legal standards and regulations
- Association recommendations
- Corporate supply specifications
- Contractual provisions:
  - Supply specifications
  - Project engineering guidelines
  - Sample plans / working templates

The bidder will be electronically informed of any changes to the contractual provisions by Mercedes-Benz AG. The bidder shall examine the content of these changes and raise any objections within a period of two weeks. Otherwise the change will be considered to have been accepted without any additional expense for Mercedes-Benz AG.

A detailed list of the Mercedes-Benz AG contractual provisions can be found in Enterprise Content Management (ECM).

The regulations and contractual provisions only list deviations and additions to the preceding regulations.

The aim of the contractual provisions is not only to standardize processes but also to support the supplier by simplifying the use of the specified systems through practically oriented, immediately usable aids and comprehensive support.

In case of contradictions in the content of the contractual provisions, the following order applies:

1. The general procurement terms and conditions, the special procurement terms and specific procurement terms of Mercedes-Benz Group AG and its subsidiaries.

2. The specifications and any associated changes and additions laid down in Volume 2 of the tender, as well as jointly reached written agreements and decisions

3. The project requirements (Volume 3) of the tender, with precedence over the Group regulations and trade association specifications, and also precedence over the legal standards and regulations provided this is permissible.

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4. The special non project-specific provisions in Volume 1.

The contents of the specifications issued by Mercedes-Benz AG, including changes and supplements, are decisive and take precedence over the bids submitted by the bidder.

The higher-level requirements generally apply. In any case, the contractor shall ensure that a decision on the design variant is made by Mercedes-Benz AG.

2.3. Invoicing and payment for products/services

The terms of payment will be determined by the purchasing department in the course of contract placement negotiations.

2.4. CE Conformity / UL-Certification / Conformity marking

The supplier is responsible for properly affixing the conformity marking required by law in the respective destination country.
The regulations and standards according to the EU Machinery Directive apply to machinery and systems brought into circulation within the European Economic Community.
The boundaries of the system are to be determined as appropriate and in conformity with the Machinery Directive (totality of machines, networking), and must be agreed with the Client. An individual conformity mark must be assigned for each autonomous system control area (PLC). The declarations of conformity must be obtained no later than the time of "Technical safety approval". All changes carried out by Mercedes-Benz AG with its employees and simultaneously communicated to the supplier as per the description in the section "Technical safety approval" must be incorporated into the declaration of conformity at no extra cost by the supplier.
If the order refers to the expansion or modification of existing systems/machinery, the supplier must renew the existing conformity mark(s) for the entire system(s) or explicitly confirm in writing that the existing conformity mark(s) will be retained unchanged and without restriction for the entire system. In case of any uncertainty as to whether a commissioned modification/conversion constitutes a minor or major change, the bidder must seek clarification between Mercedes-Benz AG and himself before the order is executed.
As an integral part of the order, the interfaces with all adjoining machines/systems shall also be examined from a safety perspective (risk assessment, risk prevention, description of residual risks) and incorporated into the conformity declarations to be issued. The risk analysis created for the machine/system by the supplier in line with the EU Machinery Directive will be made available to the Client in its entirety.

Moreover, the requirements with respect to cyber security must be taken into account, especially if the machinery is based on software or AI functions. Comprehensive technical documentation containing all machine-related information, including the risk assessment, safety measures and the declaration of conformity, is required. The operating instructions must contain all the necessary information concerning the safe use and safety requirements of the machinery.

2.5. Subcontractors, employees of the supplier

The following applies in addition to DBL 9606:

The supplier undertakes to provide the main scope of supply and services of the order using his own personnel and on his own responsibility.

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The supplier is only entitled to partly or wholly delegate the provision of products/services to a subcontractor after prior written consent/electronic consent via the Supplier Database (SDB) of Mercedes-Benz Group AG.
It is mandatory to comply with the requirements of the client with respect to the use of subcontractors.

Intended sub-suppliers must be nominated to Mercedes-Benz AG in writing before every order placement. Approval of a sub-supplier by Mercedes-Benz AG does not relieve the Contractor of his due diligence and contractual obligations. In the event of an order being placed, the sole contractual partner of Mercedes-Benz AG is the Contractor and not his sub-suppliers.

Mercedes-Benz AG shall be entitled to reject the use of individual employees of the Contractor if this is necessary for safety reasons or as a result of current or previous behavior by the person in question.

The use of vicarious agents or "hired labor" in the rooms or on the premises of Mercedes-Benz AG who have been provided to the contractor or his sub-suppliers by third parties with official permission according to Sections 1 and 2 of the German Law on Temporary Work, or in disregard of the legal regulations on the provision of work services, is not permitted.

The Contractor shall nominate the employees he intends to use in good time for the arrangement of access to the plant premises, and agree any changes with Mercedes-Benz AG in advance. Reports will only be recognized for the employees named by the Contractor. The Contractor shall ensure proper safety instruction for his employees. The orders issued by Occupational Safety at the plant shall be observed. The Contractor may only use foreign workers requiring a work permit to fulfill his contractual obligations if they are the Contractor's own employees. Another prerequisite is that these employees possess a residence permit and work permit that are valid for the area and time of the work. The Contractor will satisfy himself that these conditions are met before the employees commence work, and verify this to Mercedes-Benz AG on request. If foreign employees who are not fluent in German are used, it shall be ensured that an interpreter is constantly on site no later than by the commissioning phase, and this person must not be replaced during the assignment. Furthermore, the contractor may not use any employees of Mercedes-Benz AG.

2.6. Scope of tender

2.6.1. General Remarks

The function and the performance data of the system to be realized are described in the project-specific tender and contractual provisions (Specification, Volume 2). Illustrations are not true to scale in some cases. They are intended as an aid for costing and to illustrate the production content, however they does not relieve the contractor of his responsibility for achieving the described functional performance and ensuring compliance with all relevant regulations and guidelines.
The contractor shall deliver production equipment of simple and reliable design. The equipment shall have a low maintenance and low wear design.
He also affirms that all scopes (including those not explicitly named here) required for the provision of the described products/services are contained in his bid. The basis is all documents and data provided up to order placement.
Aids to be produced by the contractor specifically for this order and required for assembly and commissioning shall remain the property of Mercedes-Benz AG following final acceptance.
Any deviations from the specification (Volume 2) and contractual provisions are to be clearly described and summarized and assessed as a separate document accompanying the bid. Mercedes-Benz AG shall confirm its acknowledgment of these deviations by means of a signature. Services to be offered as an option shall be negotiated and ordered if necessary. If the contractor is of the opinion that the

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specifications in this tender and any change requests cannot ensure proper operation as defined in the task description, or if individual points or scopes of the tender are not clear, he is obliged to notify Mercedes-Benz AG of his reservations in writing and if possible resolve them before order placement. If the contractor fails to notify his reservations in writing, he has unlimited liability for the fault-free operation of the entire project.

The supplier will be given access to DifaNextGen system to price his bid.
The supplier is required to price the bill of quantities for hardware items and also service items made available there in full, and using the prescribed structure.
Costs that cannot be priced in enough detail using the prescribed structure are to shown as a flat rate DifaNextGen and broken down in the written quotation. Detailed reference will be made to these when discussing the specifications.

2.6.2. Green Production

In the interests of efficient use of resources, water consumption, waste, volatile organic compounds (VOC) and energy consumption for all operating modes (e.g. production, maintenance, stand-by, ...) of machinery and systems must be reduced and kept to the technically feasible minimum. Measures to reduce the generation of carbon dioxide and further decarbonization must also be taken.
Energy requirements can be reduced across all forms of energy (electricity, natural gas, operating heat, etc.) and consumables (compressed air, coolants, cooling lubricants, process exhaust air).
Every possibility for improving energy efficiency needs to be proposed and conceptually detailed.
The following are to be considered and addressed by the Contractor:

- The supply aspects relating to consumption must be shown in detail for all operating statuses of the machine/system.
- Proposals are to be submitted for the design of the system layout and procedural plan.
- The installation of measurement technology and measures optimizing consumption (measurement of consumption, transfer to energy management, potentials for CIP) must be envisaged.
- The energy consumption over the expected system lifecycle is to be calculated and (if any) compared with preceding or existing systems and the state of the art.
- The increase in energy efficiency compared to preceding or (if any) existing systems and processes is to be shown.
- The recovery of energy must also be envisaged.

If doing so results in deviations from the generally applicable requirement specifications (standard tender for assembly), operating equipment/system specifications or material approval list (including all Integra-relevant components), this must be indicated in the calculation as a separate item.

The energy efficiency of the scope of supply represents one of the decision-making criteria considered when selecting the supplier.

In cases where Mercedes-Benz AG specifies reference values, the aim shall be to achieve planning and measurement values that are lower than the specified values of the reference system. The measurement values shall be provided by the time when final acceptance of the machine/system takes place. If no reference values are provided, the supplier is obliged to identify potential savings in relation to a standard machine/system designed in accordance with the current requirement specifications. If neither reference values nor standard machine/system values are available, the machine/system shall be configured for optimum energy efficiency within the overall context of the manufacturing facility.

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The options must always be shown separately with pricing, and with the respective potential savings calculated for 5400 work hours per annum in kWh/a (x 3x75 h shifts for 240 working days), or component-specific in kWh/component part.

2.6.3. Delivery Performance

All deliveries shall be in brand new condition based on the generally approved technical practice.

The use of materials containing silicone, asbestos, PVC or Teflon and substances that attack layers of paint is strictly prohibited. For important reasons, an exceptional approval can be requested in the case of materials containing Teflon.

The work to be performed shall be carried out correctly and professionally. The work must be performed in accordance with the project specifications (Volume 3).

If there are areas with a risk of explosion, the corresponding regulations shall be observed. In the case of equipment in such areas (Zone 0, Zone 1), a certificate from the National Metrology Institute (PTB) must be submitted before acceptance of the system by Mercedes-Benz AG. For Zone 2, the submission to Mercedes-Benz AG of a manufacturer's declaration for the components used is sufficient.

2.6.4. Component Data

Current component data shall be digitally provided to the supplier at the times specified in the schedule.

It is planned that defined component statuses will be installed for the production tests. The resulting modification packages will be made known in good time for the production tests/S-LOT's and from Job No. 1, and sent to the bidder as component data sets. Here too, a reconciliation must be made with the modification status of the production facilities so as not to jeopardize the safe execution of the production tests. The data handling required for the data exchange (comparison of old with new version) and the indication of necessary measures is to be handed by the bidder and shall be included in the bid price as part of the order.

2.6.5. Scope of tender

The key components of the quotation are listed below:

- Project proposal
- Design
- Documentation
- Production
- Delivery
- Installation and
- Commissioning of the complete manufacturing facility, i.e. of all components, specific test equipment, services and programs required for operation of the system. All ancillary services and verifications shall be contained in the bid, including those which are necessary for an operational manufacturing facility but have not been specifically mentioned in the specification.
- Regular reporting of the project status at the installation location by the bidder.
- Integration of components provided by Mercedes-Benz AG into the system documentation.
- All the necessary personnel resources to achieve the quality goals of Mercedes-Benz AG (product quality, cycle time and volume).

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- The process FMEA for the project shall be taken into account by the bidder. On the basis of this, a Design FMEA is to be carried out by the bidder for critical parts of the facility defined by Mercedes-Benz AG after order placement. The results of the FMEA shall be made available to Mercedes-Benz AG.
- Support during test planning (focus on test equipment planning)
- Provision of the replacement parts package for assured availability up to acceptance according to German Civil Code.
- Training and instruction of operating and maintenance personnel (mechanics, electricians and operators) in the production systems. The planned time required for training and instruction must be shown.
- Training requirement for innovative components or those not included in the materials approval list, also new production procedures.
- Training that may become necessary beyond the extent of the planned instruction in the view of the bidder must be offered separately.
- Start-up accompaniment with own personnel of the supplier up to "Process approval of operational system" (see 2.6.10.).
- System acceptance according to Planning Guideline in Volume 3.
- For new PLC-controlled systems, all activities and procedures are to be simulated in advance in a suitable simulation program (e.g. Delmia, Process Simulate...) and virtually commissioned with the software "RF::Suite" (ECM Volume 3, VIBN). Created offline programs and simulations must be made available to Mercedes-Benz AG.
- Up to technical safety approval, the bidder will clean his equipment area at regular intervals, if necessary by himself.

The data handling required (reconciliation of old with new version) and the indication of necessary consequences shall always be part of the basic order.

- Larger systems must be of modular construction (module size = transportable equipment areas). The production system is completely constructed and commissioned on the bidder's premises.
- Project-specific deviations and changes by the bidder must be separately identified in the bid, and agreed with Mercedes-Benz AG before contract award with reference to the relevant section in the specifications. If the contractor fails to draw attention to such contradictions, the client can also demand a free replacement or modification later up to final inspection. The Contractor shall then carry this out 4 months at the latest after being requested to do so in writing. (see 2.6.1.)
- Shift accompaniment (for the period after the milestone "Process approval of operational system"), broken down into mechanics, electricians, SPS programmers, robot programmers per week for the periods "1st shift operation", "2nd shift operation" and "3rd shift operation".

The following shall be indicated as options:

- On request, an offer for a service/maintenance contract for the required system is to be prepared.

2.6.6. Presentation of Bid

When preparing the cost structure and bid, the Client requires the use of digital tools, e.g. DIFA NextGen (DNG). The bidder untertakes to use the required digital tools.

The bidder accepts that the use of digital tools may make a multi-factor authentication necessary.

Costs must be assigned to the individual production areas or model series variants. The detailed description can be found in Volume 2.

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The bid shall be structured to meet the requirements of Mercedes-Benz AG, which can be derived from the cost sheets (ECM Volume 2) or from the tables directly provided in DIFA NextGen.

- All prices must be shown in euros [€].
- The bid must be in the project language.
- Description of the structure, process and function of the production facilities.
- Layout with cross sections.
- Listing of the planned sub-suppliers with respective supply content and cost share in [%].
- Information on the total air consumption (cu. m/hour compressed air), coolant requirement (cu. m/h), electrical connection figures and consumption figures.
- Time schedule with confirmation of Mercedes-Benz AG milestones.
- Number of robots indicating types of robot, area and number of direct personnel (depositors, manual activities to be shown separately.
- The number of personnel envisaged by the bidder over the different project phases.
- Description of the instruction and training concept
- Alternative bids shall be clearly discernible as such and submitted as separate attachments to the basic tender.
- Alternative bids shall be technically comparable to the basic tender.
- The bid must include an organigram showing the project structure and cross-sectional functions.
- In the case of multiple awards (award of several projects at the same or different locations), uniform technical execution must be assured. Deviating versions or the use of different components must be identified as part of design approval and approved by Mercedes Benz AG.

2.6.7. Notes and Explanations on the Bid Content

If expanded scopes of delivery and services are agreed between the bidder and Mercedes-Benz AG, they shall be included in the bill of quantities and if necessary in the layout of the basic bid. The resulting sum represents the basis for all recalculations with regard to the additional or reduced scopes. If the scope of supply is increased, a supplementary bid confirmed by Mercedes-Benz AG is necessary. Any discounts or rebates also apply to additional scopes.

2.6.8. Personnel Requirement

Determining the number of personnel and their capacity utilization is a matter for Mercedes-Benz AG, and is the basis for system concept. Should it become apparent after contract placement that an insufficient number of personnel was planned by the Contractor for the scope of the order, Mercedes-Benz AG shall have the right to demand an adjustment. All improvements shall be specified in mutual agreement with Mercedes-Benz AG and can be both process optimizations and a higher degree of mechanization according to the technical standards defined in the specification. The costs incurred for this will be borne by the Contractor. An improvement shall be technically justifiable and shall be feasible when integrated into the manufacturing facilities without incurring any impairments to the production program.

2.6.9. Operation of the System

From the start of production activities to technical safety approval, the Contractor is the operator of the system. For this purpose the Contractor shall provide personnel with the necessary qualifications to accompany each shift.
The Contractor shall provide personnel to accompany the startup after technical safety approval. These must be present during the working times relevant to production.

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This ends with the "Process approval of operational system", or when the system substantially meets the prescribed goals and this is confirmed by the Client.
Precise agreement on this is always necessary.

The startup backup is intended to ensure that the planned production program can be adhered to in terms of timing and quality. As experience shows that systems are not yet sufficiently fault-free and reliable during this period, it is essential for the Contractor to provide an adequate personnel capacity to remedy any malfunctions or problems that occur.
Before the technical safety approval, this backup will also include operating and technical employees of Mercedes-Benz AG, and the time period will be regulated by a trial operation agreement.

Shift accompaniment will be billed according to the costs actually incurred.
This option will be taken if the Contractor has shown that the system achieves the prescribed targets and quality levels, e.g. by cycle time and availability verification in the iPortal, vPFUs etc., and is substantially fault-free (process approval of operational system), but Mercedes-Benz AG expressly requests their presence.

2.6.10. Utilization of Mercedes-Benz AG Services

With the exception of those expressly provided for, the scope of supply and services indicated in the bid contains no services whatsoever by Mercedes-Benz AG. The use of Mercedes-Benz AG employees for performance under the contract is prohibited. Should the Contractor request support from Mercedes-Benz AG to remedy faults, the these services by employees of Mercedes-Benz AG will be invoiced or offset against any remuneration claims of the Contractor. By agreement with the personnel of Mercedes-Benz AG, the Contractor has the option but not the right to commission mechanical rectification work (necessary during the commissioning phase) and to obtain replacement parts from the inventory of Mercedes-Benz AG. The prerequisite for this is invoicing via a Mercedes-Benz AG accounting number. Replacement parts taken from the client's inventory cannot be returned to Mercedes-Benz AG.

2.6.11. Site Inspection

It is absolutely essential that a site inspection be conducted prior to the submission of bid for the evaluation of the physical hall, assembly and transport conditions for a potential measurement. Any subsequent demands based on unawareness shall not be accepted.

2.6.12. Time Scheduling

It is essential to adhere to the Mercedes-Benz AG scheduling milestones when realizing the system. The contractor shall create a detailed project schedule based on these milestones as an obligatory component of his bid.

2.6.13. Alternatives

In addition to the bid for the production concept proposed by Mercedes-Benz AG, the bidder is expected to offer a more efficient and unit cost effective production concept as an alternative, taking into account the functional description. The alternative production concept must not be at the expense of production time. Alternative bids shall be clearly discernible and submitted as separate attachments to the basic bid.

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2.6.14. Working Aids

All equipment, materials, measuring devices and working aids, such as tools, lifting platforms, safety systems, etc. required for carrying out the work on site are to be provided by the bidder.

2.6.15. Additions to the Scope of Quotation

The details given in the bill of quantities may not be changed. If further explanations or additions are necessary for individual items, these must be shown separately in the bid. Other forms of tender are not permitted. Mercedes-Benz AG reserves the right only to consider bids that are complete with respect to the above when awarding contracts. Bids subject to reservation may also be rejected.

2.6.16. Additions to the Offer and Change Management

There is a general obligation of the Contractor to inform the Client of cost-relevant changes. Any increases and decreases in the order volume shall be documented on change sheets. These sheets shall be signed by both parties and are not valid until they have been signed. Services which the contractor provides without the agreed change process of the client shall not be paid for.
The extra/reduced costs will be settled as per the individual positions of the bid. Mercedes-Benz AG shall be informed of any extra and reduced costs in writing within one week, together with the reason. Ancillary costs (administration, secretarial services, backup services, office supplies plus preparation and follow-up) form part of the tender and shall not be invoiced separately. The valid Mercedes-Benz guidelines apply to travel costs.
The generally valid change management process with the change process and change documents is described in Volume 3, Chapter 67.

2.7. Rights of Ownership and Use, Acquired Rights, Intellectual Property, Rights to Work Results, Copyright

The provisions relating to the various rights and entitlements are shown in the procurement and contractual terms & conditions, see e.g. Section 6 in document "NP_20_30_100 Special procurement terms for plant services“.

The procurement terms and general terms of business can be viewed in advance in the Supplier Portal, using the "DocMaster" application.

2.8. Pricing and Settlement

2.8.1. Invoicing of Additional Quantities

If invoicing based on on-site measurements is agreed, the items in the work specification shall be invoiced according to the quantity actually delivered and installed based on the individual prices offered. At a time near the provision of the services, the Contractor shall prepare the on-site measurement by providing a tabular comparison of the quantity ordered and the quantity actually delivered, as well as a plan containing the corresponding items. The client can specify the time when this is carried out. The overmeasure check shall be carried out jointly by the Client and Contractor.

2.8.2. Hourly Work

The performance of hourly work requires a contractual agreement. Mercedes-Benz AG must be informed of hourly work before it commences.

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Hourly work slips are to be submitted showing the working hours and the necessary, separately remunerated expenses for the consumption of materials, provision of facilities, equipment, machines and machinery systems, freight costs, transport and loading services and any special costs (collectively known as ancillary costs).

The basis for settlement is the previously agreed hourly rates including the associated ancillary costs. Hourly work calculations must be submitted soon after the completion of hourly work, but at the most at intervals of 2 weeks. Evidence of work that is submitted late will no longer be recognized.

2.8.3. Wage Supplements

Supplements for nighttime, weekend and public holiday work required to perform the service described must be included in the installation prices quoted in the bid. Furthermore, additional installation costs will only be recognized if unforeseeable services to be performed during working hours qualifying for supplements are explicitly commissioned by Mercedes-Benz AG.

2.9. Faults and Warranty

In his function as the supplier, the Contractor warrants the full, comprehensive and unrestricted functional capability of the scope of delivery. This also applies to all deliveries and services of his sub-suppliers and to their interaction. The contractor must independently coordinate all interfaces between his different product areas and sub-suppliers. For this purpose, he must appoint a suitable project management team that is responsible for managing and coordinating the sub-suppliers. In all cases, the Contractor will ensure that the subcontracted scope of delivery is available on time and in the contractual condition when selecting his suppliers, formulating contracts with them and by regular reporting.

2.9.1. Claims arising from Defects, Extent and Burden of Proof

The contractor shall warrant that the deliveries (including any outsourced parts used) and services he provides will remain free of defects for a period of 24 months from formal acceptance. This also applies to multi-shift operation.

The relevant legal provisions apply in the case of construction services.

The availability of all replacement parts contained in the scope of delivery of the Contractor shall be ensured by the Contractor within the warranty period, and defects that occur shall be remedied free of charge. Faults and weaknesses remedied by Mercedes-Benz AG which the Contractor should remedy at own expense according to his contractual duties can be invoiced to the Contractor by Mercedes-Benz AG if the Contractor fails to meet his contractual obligations. This is conditional upon written notification of the Contractor by Mercedes-Benz AG before the work is carried out.

Within the warranty period, the increased maintenance cost caused by faults and weaknesses will be borne by the Contractor.

The activities and timing for the elimination of defects shall be coordinated between the Contractor and Mercedes-Benz AG. The contractor undertakes to work through the deficiencies documented in a jointly created list of defects by the agreed time. Should the contractor fail to meet this obligation, the client reserves the right to commission a further supplier to work through the deficiencies. The Contractor shall bear the resulting costs. No period of notice on the part of Mercedes-Benz AG is required. The

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contractor shall be liable for deficiencies claimed within the warranty period until they have been completely eliminated.

The contractor also bears the burden of proof that his product/service is not deficient after BGB acceptance. However, this only applies if Mercedes-Benz AG produced evidence that there is no damage by external force.

If a threat of damage or injury can only be avoided with the elimination of deficiencies by Mercedes-Benz AG, the Contractor shall bear the costs resulting from this. In all such cases, Mercedes-Benz AG will seek agreement with the Contractor before work is carried out. If a deficiency cannot be remedied, or if Mercedes-Benz AG feels it cannot be expected to attempt further improvements, Mercedes-Benz AG can demand a price reduction or withdraw from the contract. Claims for damages remain unaffected. If a component fails owing to a deficiency prior to formal acceptance, the contractor shall bear the resulting costs for its replacement and restoration of the system's operating ability. In the process, the production downtime must be kept as short as possible. Should personnel not be available to the Contractor, Mercedes-Benz AG will carry out replacement of the failed component and restoration of the system's readiness for production. Mercedes-Benz AG is entitled to invoice the Contractor for this.

Repair and customer services must also be available for a short period after the warranty has expired.

All documents or specifications provided by Mercedes-Benz AG do not absolve the Contractor of his due diligence and general responsibility for the correct design, reliable functioning, compliance with legal requirements and implementation according to generally approved technical practice.

Should regulations exist that affect possible claims for deficiencies, these shall be clearly indicated in the instruction manual or technical documentation.

2.9.2. Claims Deadline

The deadline for claims by Mercedes-Benz AG is 24 months following formal acceptance.

The legal deadline of 5 years applies to construction services.

Availability of the hardware and software and maintenance of the delivered parts shall be guaranteed by the Contractor for 15 years after formal acceptance.

The statute of limitations for claims made due to a specific deficiency shall be suspended if Mercedes-Benz AG issues a written notification of defects until the deficiency has been eliminated. However, this suspension shall end three months after receipt of the written declaration of Mercedes-Benz AG that the deficiency has been eliminated or no deficiencies exist.

2.9.3. Validity

The above provisions 2.9.1 and 2.9.2 apply to all changes and additional services commissioned in conjunction with the basic order.

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2.10. Replacement Parts, Repeat Orders and Warranty

2.10.1. Replacement parts documentation

The Contractor is required to submit the replacement and wear parts list (SLE) when the respective milestones are reached.
At the earliest possible time, explicit reference must be made to all component parts with a delivery time in excess of four weeks.

2.10.2. Provision of Replacement Parts

From commissioning until formal acceptance, and at his own expense, the Contractor must provide sufficient numbers of the necessary replacement parts on site at least until Mercedes-Benz AG has sufficient supplies of the required parts in store. The possible provision beyond the acceptance point starts at the time of handover of the final valid replacement parts lists and then lasts for an appropriate test period, a suitable processing period for in-house procurement by the Client, and for the procurement duration (delivery time) for the parts.
If Mercedes-Benz AG does not procure the recommended spare parts within the appropriate period as described above, the Contractor is under no further obligation to continue making the recommended replacement parts available at his own expense.

In order to manage and maintain availability of the replacement parts on site, the Contractor must install and manage a dedicated storage facility at the location. The Contractor shall ensure access to this at any time in the event of a malfunction. He guarantees that further parts will be available within 2 days in the event of malfunctions. Exceptions to these provisions are only permissible if Mercedes-Benz AG is already in possession of the necessary replacement parts prior to commissioning and they are available for this purpose. The Contractor shall ensure in advance that this is the case for each replacement part, and have its availability confirmed in writing by the person responsible at Mercedes-Benz AG. A list of replacement parts that do not need to be provided shall be created by the Contractor in good time and countersigned by Mercedes-Benz AG. If replacement parts belonging to Mercedes-Benz AG are used, these shall either be reprocured by the Contractor by agreement or offset against the cost of the entire order. The contractor must promptly and independently carry out the free exchange of replacement parts ordered on the basis of incorrect and incomplete documentation. In particular in case of malfunctions during the warranty period, the contractor shall ensure fast, unbureaucratic assistance. Mercedes-Benz AG is under no obligation to adopt or accept replacement parts provided by the Contractor. However, the Contractor shall have the option of selling as-new parts to Mercedes-Benz AG as required.

2.10.3. Repeat Orders

If the parts in question are not produced by the Contractor himself but sourced from sub-suppliers (purchased parts), these sources of procurement shall also be detailed and the parts specified in such a way that the Client is able to place repeat orders for these parts directly with the manufacturers without risk of confusion. Costs incurred by the client in the course of a malfunction or during maintenance as a result of incomplete, incorrect or ambiguous replacement part data may be passed on to the Contractor at any time.

2.10.4. Guarantee of Supply

The contractor undertakes to supply Mercedes-Benz AG with all replacement and wear parts over a period of 10 years (for successor-flexible production systems for 15 years), calculated from the point in time of the formal acceptance in accordance with the German Civil Code (BGB). Regarding parts

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produced by the Contractor himself, he may relieve himself of this obligation if, at the time of acceptance, he presents the client with the drawings for the actual design, including those of the contractor-specific "standard parts" in the form of production-ready drawings, calculations and other technical documents of relevance to the delivery, and transfers ownership of the same to Mercedes-Benz AG. The intellectual property right to them are not affected. Mercedes-Benz AG is then entitled to use these technical documents to carry out maintenance work and to manufacture replacement and wear parts free of charge. For replacement and wear parts which the Contractor procures as purchased parts from other manufacturers or sub-suppliers (see above) (DIN and standards parts of the supplier/Contractor), the Contractor shall ensure that parts of this kind are available at least during the specified period of 10 years (for successor-flexible production systems for 15 years) following the formal acceptance in accordance with the German Civil Code (BGB), via the contractor or the parts manufacturer. Otherwise the contractor shall bear the costs of modifications and adaptations to the system for the successor component. Mercedes-Benz AG is free to carry out the modification itself and invoice the Contractor, or to commission the Contractor to do so at his own expense. The contractor may only use parts and components as purchased parts that are approved in the specifications and parameters, specifically the BQF (approval of procurement source) or MFL (material approval list). If this is not possible for technical reasons, all deviations must be individually approved in advance and in writing by the responsible specialist department of Mercedes-Benz AG. Components for which supplies of spares are not guaranteed for the period may not be used even if they have been approved in the DBL, BQF, standards or material approval list. Should it be necessary to use components which will no longer be available within 10 years (for successor-flexible production systems for 15 years), these must be communicated to Mercedes-Benz AG to seek approval for their use.

2.10.5. Warranty

The warranty period for replacement parts is 24 months after first use in production operation (formal approval), but ends no more than 3 years after delivery. In the case of wear parts, the Contractor guarantees a minimum operating time/service life to be defined in writing for each wear part in multi-shift production conditions. This documentation must be provided by the time of technical safety approval at the latest. If this minimum operating time/service life is not achieved, the contractor must in each individual case provide the client on request with free replacements within an appropriate period (depending on the respective parts).

2.11. Replacement Tools and Acceptance

2.11.1. Replacement Tools

All original tools must be replaced with replacement tools by the Contractor by Job No. 1. In terms of dimensions (tolerances) and installation, the replacement tools must be absolutely identical to the original tools. The replacement tools must bear the same manufacturing equipment (BEMI) markings as the original tools.

2.11.2. Acceptance of Replacement Tools

In the presence of the responsible operators and the Maintenance department of Mercedes-Benz AG, and under series production conditions, the Contractor shall verify the function of each replacement tool wherever it is used. If no deviations or malfunctions become apparent (reprogramming and adjustment), the replacement tool is deemed to be accepted. The acceptance must be documented in writing.

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2.12. Service and Maintenance

2.12.1. On-Site Maintenance Activities

During the entire commissioning phase, the Contractor shall ensure all inspection, and repair activities including the necessary parts replacement required according to his own " specifications up to the point of formal acceptance in accordance with the German Civil Code (BGB). This is intended to counteract wear caused by commissioning activities and test operation, and to maintain the new condition of the manufacturing facility. The signed confirmation that this has been done shall be submitted to Mercedes-Benz AG by the time of technical safety approval, either on the Contractor's own order documentation or, if available, on the maintenance orders produced by the maintenance system of Mercedes-Benz AG. The inspection and maintenance measures required from the standpoint of the contractor shall be compiled in the form specified in the project engineering guideline Technical Documentation as part of the project specifications (Volume 3). All maintenance measures shall be so dimensioned with regard to type and scope (required time) that they suit the actual requirements and, with regard to the requested technical availability, can be carried out by Production and Maintenance within the scope of the specified shift models. The content of the measures to be carried out shall be coordinated by mutual agreement between Mercedes-Benz AG and the Contractor. All machine and system elements (mechanical systems and control systems) shall always be designed and positioned so that they are easily accessible without special tools, and can be quickly and efficiently inspected, serviced and replaced.

2.12.2. Service and Maintenance Contract

On request by Mercedes-Benz AG, the bidder must submit an offer for a service or maintenance contract covering the tendered services.
The basis for this is the maintenance and service schedule prepared by the bidder.

2.12.3. Liability

The Contractor shall absolve Mercedes-Benz AG of all claims by third parties which arise owing to a violation of contractual or other legal obligations by the Contractor.

The Contractor shall also absolve Mercedes-Benz AG of claims for damages made by employees of the Contractor. This does not apply in the event of intentional action by Mercedes-Benz AG or if an insurance company pays for the damage.

If, systems, real estate, products or other property used by the Client are damaged during the execution of the order for reasons for which the contractor is answerable, the Contractor shall be liable to pay damages.

Before the start of installation, the Contractor is entitled to record any damage that may be found at the location during an inspection and document it together with Mercedes-Benz AG. If the contractor neglects measures that he would be obliged to implement based on the agreements reached, he shall assume sole liability for all damages arising from this neglect and shall not be entitled to demand compensation from Mercedes-Benz AG. This particularly applies to the scrapping of vehicle bodies and the disposal of contamination or pollution.

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3. Project Administration and Organization

3.1. Project Implementation

3.1.1. Basic Obligations of the Contractor

The Contractor shall execute the order assigned to him on its own initiative and responsibility. The project language is German. If the Contractor is the general contractor according to the specifications of the scope of supply and services, he has total organizational, technical, timing, commercial and safety responsibility for implementation of the order scope. If technical or organizational interfaces arise with construction services or services to be provided by third parties when executing the order, the Contractor shall be obligated to independently coordinate these with the respective suppliers company or their representatives. This particularly also applies when preparing schedules and for preservation of safety at the construction site. Immediately after being assigned the order, the Contractor shall make sure that all information in the tender is still valid. Technical data shall be checked with regard to the required functional performance. This inspection also includes on-site measurements. All coordination meetings required for the order shall preferably take place at Mercedes-Benz AG.

3.1.2. Construction Services

If the bidder discovers that execution of the project will require on-site construction services that have yet to be implemented or initiated by Mercedes-Benz AG, he shall indicate this when the contract is signed and exclude these tasks from his scope of supply. The bidder shall bear the costs in case of default.

If this is not possible, the necessary information on the scope and date must be communicated to Mercedes-Benz AG in good time before the planned completion date. Specifications required to implement construction services, e.g. foundation plans, stresses, static and dynamic influences, beam grids, floor characteristics, etc., form part of the Contractor's services and must be provided when submitting the bid. Before commencing work, the Contractor shall check the on-site construction work that relates to the scope of delivery in good time. Any deviations or defects identified shall be reported in writing to the Mercedes-Benz AG project manager for the purpose of clarification and rectification. If the Contractor recognizes that construction work on site is delayed to the extent that his own deadlines are at risk, or that additional costs will be incurred, he shall inform Mercedes-Benz AG in writing without delay.

3.1.3. Project Management by the Bidder

The project structure proposed in the bid shall be confirmed and maintained by the Contractor following order placement. In the event of the absence of a team member, the Contractor shall appoint a suitably qualified replacement within 7 days. The bidder shall appoint a German-speaking (or relevant project language) project manager who remains responsible throughout the order administration and has signature rights on behalf of the Contractor. The project manager can only be replaced following prior agreement with Mercedes-Benz AG. The bidder shall nominate qualified, German-speaking sub-project managers, site managers or installation managers and specialists for all technical and organizational aspects (e.g. quality, technical safety, electrics, software, design, machinery, simulation, utility supply, etc.). Sub-suppliers commissioned by the Contractor must appoint a German-speaking site manager to coordinate on-site activities The project manager shall act as the primary contact person for Mercedes-Benz AG, and shall be given the necessary representation authority by the Contractor.
The Contractor shall ensure that the project manager can be contacted at all times.

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The tasks of the project manager and the sub-project manager particularly include:

- Management and coordination of the overall project.
- Coordination of the design, production, sub-suppliers and construction site.
- Coordination and informing of his suppliers.
- Management and coordination of the project meetings which are held regularly on the premises of Mercedes-Benz AG with its relevant departments.
- Ongoing scheduling, time management and project transparency.
- Continuous cost control and change documentation.
- Assumption of responsibility for safety at the construction site (at least organizationally).
- Identification of alternative solutions.
- Joint evaluation of alternative solutions with Mercedes-Benz AG (including the risk analyses).
- Optimization of selected concepts.
- Coordination with all specialist departments and companies involved in the installation
- Execution and signature of approvals and acceptances
- Responsibility for documentation in the required languages

There is a general obligation of the Contractor to provide information to Mercedes-Benz AG on any deficiencies, malfunctions and faults that have become known during the performance of the contract, and on any need for action to maintain the functional availability and value retention of systems and facilities. If a milestone is not reached, the status report shall be submitted by the management of the Contractor at the installation site of the scope of delivery.

These aspects are to be recorded in the official record or list of deficiencies. All documents and files are to be in the German language.

3.1.4. Regular Reporting (Status Reports, Records and Lists of Deficiencies)

The contractor shall always prepare records of all meetings, agreements, acceptances, visits, etc. In mutual interest, these shall be countersigned or agreed upon in emails by both parties prior to distribution. If such records require amendments, this shall be indicated in writing within 2 weeks after receipt. Likewise, amendments must be redistributed within 4 working days. The minutes of meetings shall be distributed by the keeper of the minutes to the participants and the project managers of the Contractor and Mercedes-Benz AG.
Furthermore, the Contractor shall present a status report showing the current status and results of the individual planning steps and work on all assemblies at intervals of two weeks, if required also at shorter intervals. The status report shall cover at least the following points:

- Actual vs. target costs
- Time schedule
- Current layout status
- Compliance with required production and systems parameters

Shortcomings shall be jointly listed and followed up in a tool provided by Mercedes-Benz AG.

3.1.5. Steering Committee

Depending on the project size and weighting, Mercedes-Benz AG will establish a steering committee in which primarily strategic and cost-relevant decisions are made. It is expected that the steering committee will meet every 2 months. The meetings will be chaired, organized and minuted by Mercedes-Benz AG. Project documentation for these meetings and presentation material for decision-making are

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to be created by the Contractor in a number and form to be decided in advance. Transitions between various project statuses shall also be shown.

3.1.6. Scope of Coordination

The contractor shall be obligated to order and carry out all coordination efforts required for the fulfillment of its scope with other participants and affected suppliers and departments of Mercedes-Benz AG. Close coordination and involvement of Mercedes-Benz AG is required. The Contractor shall notify Mercedes-Benz AG in writing of system parts that require acceptance or monitoring by the trade supervisory authority, the technical inspectorate (TÜV) or other institutions. The Contractor shall be responsible for conducting the acceptance inspections.

The Contractor shall be obligated to contact Mercedes-Benz AG in good time prior to the start of assembly on the exact scheduling procedure and to clarify details.

If a safety coordinator is made available by Mercedes-Benz AG, he is to be involved in the project and kept adequately informed at regular, mutually agreed intervals.

The Contractor is required to participate as part of the function of safety coordination (see Sec. 6 BGV A1). If work provided by the Contractor as part of the order coincides with the activities of his sub-suppliers or third parties including Mercedes-Benz AG with respect to time and location, the Contractor shall be obligated to coordinate this with the other participants and shall ensure that no hazards arise on his part. Should there be reservations on the part of the Contractor regarding the safety of his employees or employees of Mercedes-Benz AG or other third parties, the appointed safety coordinator of Mercedes-Benz AG shall immediately be informed.

3.1.7. Layout

3.1.7.1. Layout Design

The rough concept from DifaNextGen (DNG) will serve as a basis for further design of the detailed layout in the design and detailing phase.

Creation of the (system) layout will be carried out by the bidder in the FAPLIS system.

This requires compliance with the FAPLIS-CAD guideline and the CAD standards contained therein, which can be found at www.faplis.de.

The guideline specifically regulates the technical and organizational procedure for CAD data exchange, and contains the creation requirements for CAD models and their structure. In particular, the 3D layouts must be drawn with MicroStation in the CARF layout module and uploaded to FAPLIS.

The following are to be added to the layout on order placement:

- The layout consists of 2D elements and 3D models in the dgn format. The number of workers and their working area shall be recognizable in the layout.
- The arrangement of all control cabinets, electrical and control components is to be shown in the layout. SPS, console and safety areas are to be shown in separate layouts.
- During layout planning, all required production aids shall be taken into account. Equipment areas must be kept clear of unnecessary cabinets etc. to ensure good visibility and access.
- Corresponding space requirements for the staircases to the platforms provided by Mercedes-Benz AG shall be taken into account in the system layout.
- All safety fences shall be shown with safety doors.

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• Subsequent adjustments to the actual position of system components as a result of findings obtained during commissioning must be taken into account in the layout.

• All major dimensional and load figures, foundations etc. are to be shown divided into construction technology, steel construction and supply/disposal facilities (poss. in another format such as 2D layout, fact sheet etc.), so that these can be examined by the plant planners of Mercedes-Benz AG.

- All interfering contours and workspaces are to be shown transparently in 2D and 3D.
- The utility interfaces are to be shown in 2D and 3D with CARF elements. (Building and TGA interface with FP)

3.1.7.2. Building Spaces and Interference Contours

The Contractor shall plan the production systems while taking into account the specified building dimensions, the building structure, the load container dimensions and the routes for the logistical supply of the systems with production materials.

The building area envisaged for this project will be provided by Mercedes-Benz AG as a FAPLIS download. Not all structural features and peripheral areas may be shown in it. The data set must be checked by the Contractor at the installation site.

The installations include:

- adjacent production systems
- energy lines
- building technology
- supply and exhaust air ducts of ventilation systems
- conduits for utilities
- covers of the rainwater and wastewater disposal systems

If no drawings of these installations exist, the Contractor must take measurements at the installation location. The Contractor shall be obligated to check the existing installations on site in good time to prevent collision points with the production systems. The required area for workforce measures and additional installations shall be coordinated in good time and in detail with Mercedes-Benz AG. Changes to the layout and the systems to reconcile them with the actual conditions in the design and detailing phase shall not justify subsequent demands by the Contractor.

3.1.7.3. Layout Approval and Provision

When designing system layouts, the concept must be coordinated with the rough concept produced by Mercedes-Benz AG, e.g. from DNG. Mercedes-Benz AG will approve the concept and, in the course of further planning, the position of the sub-systems with the related material flow and supply concepts.

3.1.7.4. Layout Updates

The 3D layout for production facilities must aways be adapted to the current status by the Contractor during execution of the project.
In addition to the status report, this data must be constantly available in FAPLIS, in the MicroStation-dgn format and as CARF objects. This allows the current project status to be integrated into the 3D coordination model of Mercedes-Benz AG, examined in the NavisWorks (BIM environment, **B**uilding **I**nformation **M**odeling) and checked for potential collisions.

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Changes to the concept and later layout shall be coordinated with Mercedes-Benz AG and identified by a consecutive change index and brief information on the changes concerned.

All changes made locally on site must also be documented in the CAD data record. This refers to changes such as relocation of safety fences, installations, robots and devices.

3.1.7.5. Project Control and Scheduling

As part of commissioning, the Contractor will receive a rough time schedule prepared in advance for the entire project as a binding, integral part of the contract. The Contractor undertakes to gradually increase the level of detail to procedures with a maximum duration of 4 weeks after agreement and approval by Mercedes-Benz AG. The Contractor shall be responsible for deadline management.
The contractor shall provide a schedule to the client together with the order acceptance (order placement), which is based on the milestones specified in the contract. If these conditions have been met, the milestone schedule shall become binding, provided that Mercedes-Benz AG does not object within a reasonable period. The due date of payment entitlements of the Contractor shall always be contingent on receipt of the schedule by Mercedes-Benz AG. The schedule amendments which occurred with project progress and the actual processing statuses (completed, percentage completion, etc.) shall be integrated quickly in the schedule. Mercedes-Benz AG shall receive the updated schedule on request or at the latest at monthly intervals. Relevant milestones (e.g. due dates of decisions, deadlines for completion of construction services) shall be included in the milestone schedule.
Significant or critical deadline delays may only be implemented following agreement and approval by the client, and shall be clearly marked in the schedule. Endangered deadlines which may cause a delay in the project shall be reported Mercedes-Benz AG immediately in writing. At the same time, measures shall be indicated as to how delays and dangers can be compensated. All changes and delays in the time schedule shall be documented and visualized in the schedule by the Contractor.
If the desired changes or time delays are clearly caused by the Contractor, all related additional expenses, including those of third parties, shall be borne by the Contractor. The 'costs-by-cause' principle shall apply in all cases. The agreed deadlines for all services of the Contractor are binding and shall be met. If the Contractor is responsible for a delay, he is in default. The Contractor shall be liable in case of a failure to meet the deadlines for which the Contractor is responsible. Mercedes-Benz AG shall be entitled to set new deadlines for the services of the Contractor, or to demand that the Contractor interrupts his services.

Should Mercedes-Benz AG fail to meet its deadlines, the Contractor cannot postpone his subsequent deadlines any further than the delay incurred by Mercedes-Benz AG.
As part of time scheduling, the Contractor shall prepare and keep up to date an installation schedule in which the various installation stages and their timing are represented. At the request of the client, the number of planned installation personnel shall also be shown. For conversion work on weekends, precise hourly time schedules must be prepared. If the ordered scopes are part of a more extensive conversion measure, the required partial schedules for an entire project shall be prepared and updated. Over the entire project period, the time schedule is to be prepared using Mercedes-Benz AG approved software and shall be updated using this software version until the end of the project. The schedule is generally published as a networked bar schedule. The critical path is to be highlighted. Specifications, such as column content and designations shall be taken from the specified schedule and any necessary expansions shall be coordinated with Mercedes-Benz AG and recorded in a file to be supplied with the documentation.
If deviations from specifications or joint decisions result, or if work cannot be carried out for reasons for which the Contractor is not responsible, Mercedes-Benz AG is to be informed immediately in writing with indication of corresponding reasons.

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3.2. Documents Requiring Approval

In good time before the required milestones, the Contractor shall submit the documentation according to ECM Volume 3 "–Planning Guideline" for examination and approval. The Contractor shall make particular reference to any designs that deviate from the design guidelines for urgent reasons.
The examination shall be carried out in planning and design meetings and shall be recorded by the Contractor. The examination and approval by Mercedes-Benz AG do not relieve the Contractor of the obligation to ensure the implementation of the overall system under the terms of the contract.
The changes required by the departments to ensure the standard or the specifications must still be possible without scheduling consequences.
If he fails to comply, the Contractor bears the sole risk and responsibility for any resulting on-site modifications (including costs) and the consequences of time delays.
If these documents are not submitted in good time, and if this results in the Contractor incurring costs due to subsequent change requests by Mercedes-Benz AG, these costs shall be borne by the Contractor.

3.3. Installation Manager and On-Site Administration

3.3.1. Installation Manager

Prior to the start of installation, a responsible installation manager shall be appointed who shall be continually present at the installation site from the opening of the site until at least the technical safety approval, and must not be replaced for the duration of the project. Any replacement, or the absence of the installation manager requires prior notification and consent by the specialist department of Mercedes-Benz AG.
The personnel used at the construction shall know the assembly manager and shall be informed on the application location and the project designation. This also applies to the personnel of the sub-suppliers.
The installation manager and his deputy must be able to speak and write the German language and the national language of the installation site of the production equipment fluently. The installation manager is required to provide Mercedes-Benz AG with information at the regular construction meetings or on request.

The installation manager's tasks include:
- Compliance with the regulations applicable to external companies at the installation site
- Compliance with UVV safety regulations, obligations under construction site regulation RAB 32 and with Mercedes-Benz AG requirements on the site
- Monitoring the integrity of production during the course of interventions in production-relevant systems and processes
- Monitoring and directing the installation activities in their area of responsibility
- Coordination of the interfaces with existing production systems, government authorities, other suppliers, etc. via Mercedes-Benz AG project management
- Deadline monitoring
- Coordinating deliveries of the scope of supply
- Monitoring of test runs and the commissioning process
- Assistance with approvals, acceptance and testing

3.3.2. On-Site Support Facilities

Construction site equipment (containers complete with communication equipment, changing rooms, washrooms and recreation rooms, material staging areas) shall be included in the contractor's scope of delivery (also for his sub-suppliers). Before contract placement, the location, size, delivery, extent of the facility (e.g. toilets, ...) and the available space must be agreed with Mercedes-Benz AG. Areas for the

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construction site installations and material staging areas may only be provided to a limited degree and not in the immediate vicinity of the construction site. Multi-storey containers may have to be used.
In the course of project progress, it may be necessary to move the construction site equipment to other areas. In such a case the Contractor has no claim to additional payment. This service is included in the total price if the situation was known before contract placement.
The delivery and return delivery of construction site containers, workshop equipment, etc. are at the expense of the Contractor.
The driving of containers etc. onto the plant premises requires approval. Mercedes-Benz will illuminate the construction site solely for normal pedestrian traffic (nominal illumination level approx. 20 lx). The Contractor shall be responsible for any specially required on-site lighting. On completion of his services, the Contractor must immediately clear the installation site without any express request by Mercedes-Benz AG. Access routes, storage and workplaces provided by Mercedes-Benz AG shall be returned to their previous condition when leaving.
The delivered material remains the property of the Contractor, and is therefore his responsibility, before and after the installation until formal acceptance according to the German Civil Code. Mercedes-Benz AG accepts no liability for any losses or damage. Surveillance and storage are the Contractor's responsibility. This excludes operating equipment that has received "technical safety approval".
Power and water for the site shall be provided by Mercedes-Benz AG at the plant distribution interface.

3.3.3. Work Safety

The Contractor undertakes to instruct his personnel in Mercedes-Benz AG regulation DBL 9606 (regulation for third party companies), the German accident prevention regulations (UVV) and the occupational safety and environmental protection information, and to require them to comply with these. The Contractor shall be responsible to Mercedes-Benz AG for compliance with these regulations.

Prior to initial work at the construction site, in particular involving activities or interventions in existing systems in the context of modification or expansion work, the Contractor shall ensure, in his own interest, that a risk assessment is performed between the Contractor and Mercedes-Benz AG for the work planned, including specific instruction (Section 8 Industrial Safety Act). This is for the safety of the employees of the Contractor, Mercedes-Benz AG and third parties.
In the instruction course to be arranged by Mercedes-Benz AG, e.g. using external OiRA companies, the existing, activity-related and plant-specific hazards will be explicitly described together with the necessary protective measures (e.g. personal protective equipment) and the safety regulations to be observed.
Instruction of the Contractor must take place on an order-specific basis, but at least once a year. The Contractor shall confirm receipt of the documentation and that his personnel and sub-contractors have been duly instructed.
Mutual hazards, unusual findings and additional requirements (e.g. hazardous work arising from the task in hand) must likewise be documented and agreed with the Contractor, if necessary on site.
If the work of the Contractor poses hazards for the Client, he must communicate this to the Client. In this case a joint risk assessment and action plan is to be prepared.
On-site markings and documents (e.g. work instructions for hazardous substances) must be written in the national language.

The obligations of Mercedes-Benz AG in accordance with Section 6 BGV A1 are not affected by this (site safety coordination).

For larger construction sites, new buildings or construction scopes that require a safety coordinator, Mercedes-Benz AG will provide a safety coordinator to monitor the safety regulations. He/she is authorized to give instructions to the Contractor's employees on all safety-related aspects. Otherwise,

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the instructions issued by the Mercedes-Benz AG employees responsible for industrial safety at the plant shall be observed. The instruction authority of the health & safety coordinator does not relieve the supervisors of the Contractor of their responsibility. The respective supervisors remain accountable and responsible for their employees in all cases. They have a duty of supervision in their own area.

When asked to do so by Mercedes-Benz AG, the assembly personnel are obligated to take part in training given by Plant Safety and the fire brigade and to inform themselves about safety precautions using the occupational safety brochure.

The Contractor shall be responsible for system and construction site safety.
Areas of activity are to be adequately safeguarded by the Contractor at all times by agreement with all persons involved, with respect to fire protection, environmental protection and health & safety protection. The Contractor shall manage these so that no personnel are endangered and no physical damage can occur. The materials necessary for this shall be provided by the Contractor.

Mercedes-Benz AG reserves the right to monitor compliance with the agreed protective measures and general safety specifications by random sampling, and to punish violations with permanent banishment from the site. Furthermore, Mercedes-Benz AG shall be entitled to reject the use of individual employees of the Contractor if this is justified for safety reasons or as a result of current or previous behavior by the person in question.

3.3.4. Cleanliness

Should the Contractor fail to keep his construction site clean or sufficiently clean, Mercedes-Benz AG will carry out the required cleaning itself or have it carried out by third parties at suitable intervals if a reminder is unsuccessful. The Contractor shall bear the resulting costs.
The construction site shall be swept clean before leaving.

3.3.5. Waste Disposal

The disposal of construction rubble, packaging waste and construction site waste contaminated with pollutants during and after the installation work shall be the responsibility of the bidder and is to be included in the price calculation of the basic order. In the case of special category waste, the Contractor must provide Mercedes-Benz AG with written disposal certificates from authorized companies. If the Contractor does not fulfill his obligation to dispose of the waste, Mercedes-Benz AG will have the disposal work carried out by a specialist company at the Contractor's expense.

3.3.6. Notification of Third Party Activities

To ensure the smooth organization of work on weekends, public holidays or at night, the personnel of the contractor provided for this purpose shall be reported in each case to the responsible department of the plant at the installation site or to the responsible plant security staff of Mercedes-Benz AG in good time so that this work can be approved. The Contractor is responsible for filing any applications required in connection with this work with the trade supervisory authority. The Contractor shall comply with this obligation in good time and without being asked to do so, otherwise Mercedes-Benz AG shall have the right to reject this work without any claim to damages by the Contractor. It is particularly important to report the Contractor's employees in good time if working on weekends. Mercedes-Benz AG reserves the right to reject work of this kind for special reasons without any claim to damages by the Contractor.
The Contractor is obliged to obtain information about the reporting process and procedure at the installation site so that a smooth operation is ensured.

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3.3.7. Existing Installations

The Contractor must assume that the installation and commissioning work will be carried out in an operational production environment. Production at the Mercedes-Benz AG plant must not be impaired or interrupted by the work. The Contractor must also take into account that during the installation work, other manufacturing facilities are simultaneously being realized which could make the installation work more difficult in certain areas. To minimize mutual obstructions and necessary work interruptions, the Contractor shall ensure sufficient coordination with those involved. Any intervention in existing systems shall only be permitted with the approval of Mercedes-Benz AG. Existing installations that are in the way of the contractor's installation activities may not be removed or modified without prior authorization. Work of this nature shall be carried out exclusively according to Mercedes-Benz AG instructions. The Contractor shall inform Mercedes-Benz AG in good time of the need for any such intervention in existing systems, so that the required measures can be initiated. The Contractor shall ensure that the existing systems are returned to an operational status following the intervention. If no trial operation is carried out before the start of the next shift in the presence of Mercedes-Benz AG employees (at least at master tradesman level), the Contractor's employees commissioned with the change shall be present at the start of the shift in order to eliminate any problems that may arise. If parts of existing manufacturing facilities need to be dismantled by the Contractor, they shall always be dismantled in such a way that these parts can be reused. The contractor shall clarify the need for reusability with Mercedes-Benz AG. To the extent possible, dust-creating work shall generally be performed outside of the building. Unavoidable work within the building necessitates the use of dust protection measures, such as enclosures or wet cleaning etc., which must be coordinated with Mercedes-Benz AG project management in each case.
Materials containing asbestos may have been used in all older buildings. Prior to any demolition/conversion or addition to existing buildings, it is essential to consider the age of the building and involve the specialist departments for construction/conversion work and Facility Management. The relevant health & safety department is the point of contact for location-specific aspects.

3.3.8. Interfaces with other Areas

The Contractor shall take any interfaces with other areas into account in his plans and ensure the exchange of information. He shall ensure that the interfaces with other areas are defined exactly, and adapted to each other and tested during commissioning. The Contractor shall carry out a collision test with the areas concerned or existing equipment/installations for his scope of delivery (in particular the installation scopes). These collision tests shall be conducted with on-site measurements and inspection if necessary. If the Contractor brings functions or components into a manufacturing area that has interfaces with other areas, he shall ensure that the functionality of the affected areas is not impaired. If additional costs result in the areas disturbed by the Contractor, the "costs-by-cause principle" shall be applied. In cases of dispute, the decision lies with Mercedes-Benz AG. There is an obligation for notification and coordination in good time. In the event of interface problems (up to and including the formal acceptance in accordance with the German Civil Code (BGB)), the Contractor shall be obligated by Mercedes-Benz AG to use appropriate aids (such as records, measuring equipment etc.) to demonstrate that his contribution to the interface has been provided as agreed and that it functions properly.

3.3.9. Verification

Mercedes-Benz AG reserves the right to inspect the quality of the services, also outside the planned approvals/acceptances, either in full or on a random sample basis. Should Mercedes-Benz AG find significant or frequently repeated shortcomings, the (additional) cost to Mercedes-Benz AG of providing care and supervision shall be invoiced to the Contractor. Examples of this include:

- devices not measured or measured incorrectly

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- faulty programs
- inadequate execution of system and control components
- system design by the Contractor or his sub-suppliers:

Visual and function checks on masks, dialogs and lists /in-house tests on Contractor or sub-supplier premises

- inspection of electrical and mechanical installations
- interface configurations
- documentation

If malfunctions occur during production which are the result of serious errors made by the Contractor, Mercedes-Benz AG shall also invoice the additional expense for eliminating these malfunctions.

3.3.10. System Installation on Contractor Premises

Mercedes-Benz AG requires the installation and operation of the system components at the Contractor's plant before delivery. Employees of Mercedes-Benz AG shall be given the option of assisting the system installation at the Contractor's plant. This participation is informal in character. The Contractor cannot expect any major part of the work to be performed by employees of Mercedes-Benz AG. The responsibility for the functionality, safety and compliance with deadlines of any work taken on remains with the Contractor. If the order covers several identical systems, the Contractor shall install at least one complete, operational example system on his premises (by agreement with Mercedes-Benz AG).

3.4. Instruction and Training

3.4.1. Instruction of Mercedes-Benz AG Employees

Instruction shall always be included in the scope of delivery. It shall be completed before the technical safety approval . It is irrelevant whether it is a new delivery, modification, extension or removal. It includes instruction and familiarization with operation, fault analysis, fault elimination, repairs and preventive maintenance. Detailed instructions regarding the function and configuration shall be provided for the software scopes. The scope of the instruction shall enable the independent operation of the plants, machines and systems by Mercedes-Benz AG employees. For effective familiarization of maintenance and operating personnel, cooperation by the commissioning specialists of the Contractor is an essential requirement during commissioning (documents, documentation, questions, explanations, etc.). The instructions shall be given using comprehensive written documents relating to the system. Each participant shall be provided with a complete set of digital or written instruction material. This material shall be suitable for subsequent independent instruction by Mercedes-Benz AG. The instruction shall take into account the "users and operators" and system support personnel in all work shifts. The dates shall be coordinated with the client and their correct execution shall be recorded (contents and participants).
The instruction shall cover the entire extent of the supplied system or conversion.
8 weeks before the start of instruction, a training schedule with dates and duration of the training measures shall be provided to Mercedes-Benz AG for confirmation by the departments involved (Production, Maintenance ...).
The participant list shall be signed by the participants. The duly completed and signed participant lists shall be handed out with one set of training documents in each case as verification of proper implementation to the production and maintenance supervisors or their representatives at the installation site before the start of trial operation.

It shall be structured according to:

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Installation

- Installation of the system
- Function
- Safety devices and residual risks
- Control and operation
- Maintenance and inspection
- Troubleshooting and repair

Training content:

- Safety provisions, safety equipment in the system
- Behavior when entering systems
- Behavior in emergency situations
- Basics and operating principles of the production and process technologies and processes used
- System overview and function
- Conformity with intended use
- System operation and programming
- Process data analysis and process optimization
- Simulation of possible malfunctions, fault identification
- Performance of maintenance, repair and servicing work
- Performance of calibration and adjustment work
- Explanation of all relevant plant and function diagrams
- Structure of the operating system, the programming software and the application programs
- Interfaces with superordinate control systems and neighboring systems
- All functions of the system control console
- All functions of the system visualization at the system
- All system-wide functions

The Contractor shall be liable for all damage caused by the Mercedes-Benz AG personnel to be instructed and/or already instructed by him as a result of inadequate or incorrect instruction/training or inadequate supervision by the Contractor. This liability covers the period from initial commissioning until final acceptance of the system.

3.4.2. Training

For all new components that are not contained in the project-specific material approval list, the required training cost/effort shall be specified by the bidder and already taken into account as part of the basic order prior to approval by Mercedes-Benz AG (see 2.6.5. Scope of Tender)

Training of the operating and maintenance personnel must be completed before the technical safety approval. All the required written documentation forms part of the bidder's scope of delivery.

The names of all training participants shall be recorded. This record serves as verification for final acceptance. Dates and training activities shall be agreed with Mercedes-Benz AG. 8 weeks before the start of instruction, a training schedule with dates and duration of the training measures shall be provided to Mercedes-Benz AG for confirmation by the departments involved (Production, Maintenance...). The training measures shall be specified by the Contractor, and the necessary qualifications to be already possessed by Mercedes-Benz AG employees must be shown. The training shall be given using comprehensive written training documents relating to the system, and include theoretical and practical training of the personnel. Each participant shall be provided with a complete set of digital or written training material . This material shall be suitable for subsequent independent

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instruction by Mercedes-Benz AG. All training preparations, training documents and implementation at the corresponding plant, including all ancillary costs, shall be taken into account in the scope of the tender. The duly completed and signed participant lists shall be handed out with one set of training documents in each case as verification of the proper implementation to the production and maintenance supervisors or their representatives at the installation site.

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4. Acceptance Process

In the course of acceptances and approvals, the Client reserves the right make use of digital tools (e.g. apps such as App4Realease or similar).

The bidder accepts that the use of digital tools may make a multi-factor authentication necessary.

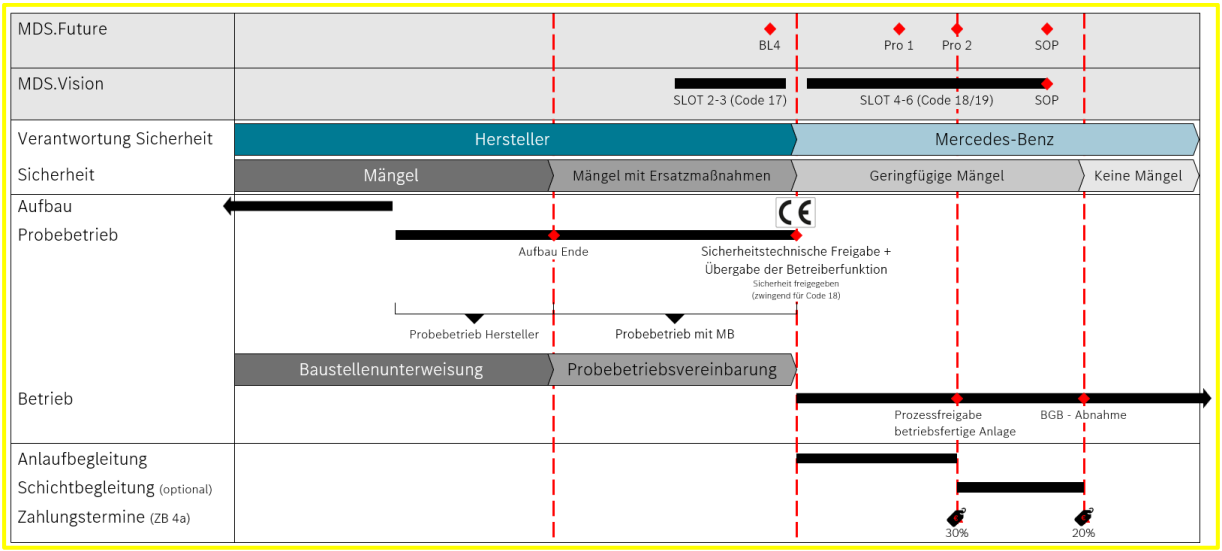

Fig. 1: Acceptance process from technical commissioning

4.1. Design approval

The design approval represents the approval from the Mercedes-Benz AG perspective of the contractor's design documents for the production of the systems/machines, and expresses the consent of Mercedes-Benz AG to the technical solutions and concepts chosen by the Contractor. For conditions and approval criteria, see Planning Guideline (Volume 3). Documentation shall be in the form of written records.

4.2. Shipping Approval

Shipping approval takes place at the bidder's site. Mercedes-Benz AG will check the requirements, functionalities and criteria laid down in Volume 2 in accordance with the Planning Guideline (Volume 3), as well as the agreed content from the planning and design meetings. The shipping approval shall be documented in the subsequent acceptance report including a list of deficiencies. The shipping approval confirms that the system may be shipped to the receiving plant.

4.3. Technical Commissioning (Trial Operation by the supplier)

The technical commissioning phase shall begin after the completion of all installation work at the location, and end at the time of technical safety approval. The starting point shall be decided between the Contractor and Mercedes-Benz AG.

The supplier is responsible for the technical commissioning!

This phase shall not constitute a technical safety approval and must not be confused with commissioning by the Contractor, but is a preparatory phase preceding this.

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Trial operation by the supplier as part of the manufacturing process comes before the transfer to the operator. This means that the necessary adjustments can be made without the machinery having to conform to the EU Machine Directive.

However, other protective measures must then be taken to comply with accident prevention regulations together with the Health & Safety Act and industrial safety regulations.

The Contractor shall operate (verify daily production, remedy faults, instruct the responsible maintenance personnel in changes, improvements and fault rectification) the system until technical safety approval, using his own specialist personnel (production maturity in trial operation). Programming and service devices shall be provided by the Contractor. During the commissioning process of the manufacturer , training shall be given to the responsible maintenance and operating personnel. If not all of the intended safety measures or devices are in place during commissioning by the Contractor for the purpose of trial operation or trial production, especially if employees of Mercedes-Benz AG are involved, or conformity has not yet been verified, a written agreement for trial operation (see ECM B3.002.04.27) shall be made between the Contractor and Mercedes-Benz AG.

If the described case occurs, a risk assessment acc. to the Health & Safety Act and an instruction manual adapted to the function test (in particular, description of the additional hazards and the required behavior) shall be produced by the Contractor and written proof shall be provided that it has been used to instruct the employees of Mercedes-Benz AG as part of the training planned for the trial operation phase. The Contractor shall notify Mercedes-Benz AG of the completion of his technical commissioning in writing. The Contractor shall then grant Mercedes-Benz AG an adequate period in which to inspect the system. After this, the lists of deficiencies created during preparations for technical safety approval of the system shall be handed over to the Contractor by Mercedes-Benz AG.

The deficiencies relevant to technical safety approval shall be remedied by the time of technical safety approval at the latest. Elimination of the deficiencies shall be accepted on site by Mercedes-Benz AG.

Chapter 6 "Miscellaneous and Appendix" describes the responsibilities and handovers to Mercedes-Benz AG if no contrary project-specific arrangements are made.

4.4. Technical Safety Approval and Transfer of the Operator Function

Between the Contractor and Mercedes-Benz AG, the technical safety approval documents that the ordered scope of delivery has been delivered in keeping with the milestone and that all safety-relevant criteria in the Planning Guideline have been met. To this end, an approval meeting will take place with a local inspection and discussion of any deficiencies.

All technical safety deficiencies must be remedied before the technical safety approval.

By agreement with Health & Safety, any minor safety-related deficiencies can be remedied after the technical safety approval.

Any other points in the list of deficiencies must be remedied by the time of formal acceptance acc. to BGB. The approval record, with a list of deficiencies if any, shall be signed by the Contractor or his appointed signatory and Mercedes-Benz AG, if desired using digital tools.

The technical safety approval signals the transfer of the operator function to Mercedes-Benz AG, and therefore operational status with the resulting requirements according to the EU Machine Directive.

Should the machine/system not yet achieve the performance data required in the specification at that time (e.g. cycle time/availability...), the Contractor shall ensure that these are met by the specified milestones.

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After the technical safety approval - with due notification of the Contractor - Mercedes-Benz AG may make minor changes (acc. to the EU Machine Directive) to the scope of delivery that go beyond the scope of the order. These changes may be made by its own employees or by third parties. The Contractor is prepared to document these changes by agreement.

Conformity marking / technical safety approval is a requirement for the production of near-series or customer-ready vehicles (vehicle code 18 or 19).

4.5. Production Tests (Pro) / Series Production Lot (S-Lot)

By means of production tests, the Contractor and Mercedes-Benz AG shall jointly and systematically determine the extent to which the production facilities and products to be produced already reach their target status at the prescribed times. This systematic approach ensures that the project target is achieved on time and with the required quality. For conditions and approval criteria, see Planning Guideline (Volume 3) as part of the basic contract. Any deviations or deficiencies that occur are registered and logged, addressed, and provided with deadlines.

4.6. Process approval of operational system

The "Process approval of operational system" will at the latest take place with Q-approval unless otherwise agreed.
Between the Contractor and Mercedes-Benz AG, the Process approval of operational system documents that the required cycle time, availability and quality levels have been verified. An approval meeting shall take place for this.
The approval record and any list of deficiencies shall be signed by the Contractor and Mercedes-Benz AG.
If payment terms 4, 4a or 4b have been agreed on contract placement, the "Process approval of operational system" entitles the Contractor to present the second partial invoice.

If the machine/system already meets the requirements for technical safety approval, the invoice can also be presented at this milestone.

4.7. Final acceptance

Final acceptance according to the German Civil Code (BGB) is an official procedure. The final acceptance (according to the German Civil Code) documents that the ordered scope of delivery is fault-free in all aspects and meets all the required criteria. For conditions and approval criteria, see Planning Guideline (Volume 3).

Mercedes-Benz AG request the acceptance according to German Civil Code. This shall be followed by an acceptance meeting in which the supplier presents and hands over verification that the prescribed conditions and approval criteria have been met.

Mercedes-Benz AG will compile a written acceptance protocol and, if necessary, a list of defects, rectification deadlines, and a specification of a financial amount retained.

If the faults are not rectified by the deadlines specified by Mercedes-Benz AG, Mercedes-Benz AG has the right to correct the faults at supplier cost and to deduct the resulting costs from the final settlement.

Once acceptance is awarded as per the BGB, the supplier is authorized to submit its final invoice.

If the supplier is not able to fulfill the requirements to prove production capability, or if acceptance is refused for other reasons, the tests are repeated after the necessary optimization measures have been carried out and the contractual parties negotiate a suitable new acceptance and payment modality.

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In this case, Mercedes-Benz AG reserves the right to charge the supplier for any costs arising from each further acceptance date. Mercedes-Benz AG reserves the right, throughout the acceptance procedure and the limitation period, to check that the machine / system complies with the requirements specified in the 9th Equipment Safety Law (GSGV) §§ 2, 3, 4 and in the EC Machinery Directive. This shall involve a visual check by Mercedes-Benz AG or third parties.

4.8. Technical Documentation

4.8.1. Form of Documentation

The technical documentation shall be in accordance with project requirement "Technical Documentation" in Volume 3 and shall be submitted in time for the required milestones. The data storage medium provided by the Contractor with the final version of the technical documentation must have a minimum storage life of 30 years, however this does not apply if the DEEP process is used (see 5.8.3.). Before the start of design work, the bidder shall conduct a coordination meeting with the Mercedes-Benz AG personnel responsible for technical documentation, in which the structure and form of technical documentation will be explained in the context of the contractual provisions. If the bidder fails to prepare the technical documentation, it is incomplete, late or fails to meet the agreed contractual provisions, the cost and effort required to produce it shall be invoiced to the bidder on the basis of the usual hourly rates for engineers. The form and content of the technical documentation received will in all cases be checked by Mercedes-Benz AG. In case of deviations from the agreed project standard and scope of delivery, the bidder is required to rectify these as rapidly as possible. If significant deficiencies are once again discovered after representation , rectification work shall likewise be invoiced to the bidder on the basis of the usual hourly rates for engineers. Mercedes-Benz AG shall be afforded sufficient time for checking and analysis. The technical documentation in its preliminary version forms part of the technical safety approval and in its final version a precondition for acceptance. Late or incomplete submission of this documentation may delay or impede acceptance. Changes and updates to the individual documents that arise as the technical documentation is compiled during the course of the project must be uniquely identified as such for the respective delivery to Mercedes-Benz AG. They must also be presented in the form of a change log for each document and as an overview detailing documentation versions for the complete technical documentation.

4.8.2. Incorporation into Existing Documentation

Insofar as the assignment concerns the modification or expansion of an existing production facility, the Contractor shall obtain the existing technical documentation from Mercedes-Benz AG no later than when the technical documentation is created, so that he can incorporate the documentation of his scope of delivery into the existing documentation or expand on it. The changed content shall be marked as specified in the project requirement "Technical Documentation", and the overall documentation presented in record able form according to the prescribed structure and format. In this case, changes that have not yet been revised and changes only entered in handwriting by Mercedes-Benz AG personnel on previous occasions shall be included free of charge. If necessary, inspection log books and acceptance test certificates must be submitted. There is no guarantee that this documentation is identical to the technical documentation issued by Mercedes-Benz AG. In his own interest, the bidder is advised to reconcile the documentation with the circumstances on site. Depending on the order content and after order placement, the Contractor shall receive data storage media with overviews/layouts relating to the tender (if available). The overview, installation and foundation plans etc. shall be continually adapted to the respective status by the Contractor when carrying out the project. On request by Mercedes-Benz AG, but at least monthly, data storage media are to be submitted so that the current project status can be referenced in the building plans of Mercedes-Benz AG. Layout and overview diagrams for factory planning and for mechanical and electric designs are to be

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implemented according to the corresponding CAD project engineering guideline of the project standard. Any conversion of data of other, non-approved systems into the formats of the systems prescribed according to the CAD project engineering guideline is not permitted. During data transfer, the data protection provisions of Mercedes-Benz AG shall be complied with.

4.8.3. Documentation in DEEP

DEEP (**D**igital **E**quipment **E**ngineering **P**ortal) is the central application for the administration and checking of technical system documentation, as well as the processing of bills of materials incl. the parts selection process.
Furthermore, DEEP is the central point for parts/material/special approvals.

Access to DEEP can be requested via the Mercedes-Benz Supplier Portal.

In the documentation meeting, the delivery milestones and scope of documentation will be determined and recorded in DEEP.

The supplier shall prepare the technical documentation for the system, and attach the necessary documents to the structural elements of the bill of materials or materials in SLE.
When uploading to DEEP, the structured bill of materials and document items are copied to DEEP.
This makes it unnecessary to transfer data by SWAN/hard drive/CD/DVD.
The structured bill of materials serves as the basis for the replacement parts ordering process, the examination process for technical documentation and importations into the SAP iPro system.

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5. Key Figures

5.1. Calculation of Technical Availability

Technically integrated availability is verified via the Integra portal.

In the case of systems without a connection to the Integra portal, availability is verified on basis of the following calculation.
When caculating availability, the technical downtimes of the system are related to the processing time (not operating time) of the system:

$$\boldsymbol{Availability\ P = Q * \frac{TE}{Q * TE + TS} * 100\%}$$

T_S	Technical downtimes of the system
Q	Number of vehicles processed in the system during the period in question. In the case of systems that do not process all vehicles, only the actually processed vehicles are counted.
T_E	System cycle time defined in the tender
$Q \cdot T_E$	Processing time in the period in question

In exceptional cases the template in ECM, Volume 3: B3.009.00.01 (with operating time) can be used if this is agreed in advance.
The result is the maximum permissible mean downtime duration with which the required availability can be achieved:

$$\boldsymbol{Downtime\ TS = Q * TE * \left(\frac{1}{P} - 1\right)}$$

5.2. Plant Utilization Time

The machine usage time is the time during which the machinery and equipment is theoretically available for production.
This does not include time for breaks or regular communications/reporting.
Likewise not included are the time blocks required for maintenance and inspection that may need to be carried out as a preventive measure on a shift basis, daily or weekly in order to maintain the production capability of the systems and take place outside the production times.
This shall be taken into account in the design of the system.
The operating time is the sum of processing time, technical system downtimes, organizational downtimes and transition times.

5.3. Transition time

Transition times are periods within the operating time in which

- the system is idle because the upstream system cannot provide a bodyshell for processing
- the system is occupied with a completed bodyshell that cannot be passed on to the downstream system.

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5.4. Downtimes (SSZ)

When calculating downtimes, a distinction is made between technical and organizational downtime. Technical downtimes are times when the system is unavailable owing to unscheduled repair or maintenance work in production.

The following downtimes are not taken into account when calculating availability:
- Downtimes occurring during break times, i.e. outside the system operating time
- Downtimes occurring at times when no vehicle is being processed (during transition)
- Downtimes occurring during the passage of vehicles that are not being processed (in systems that do not process all vehicles).

Scheduled maintenance work is to be performed outside the operating times.
100% availability is achieved if excluding the effects of transitions, no technical downtimes occur during the system operating time.

Technical system malfunctions
The following technical system malfunctions are taken into account as appropriate:

- electro-mechanical malfunctions, e.g. robot faults, positioning faults leading to line stoppage.
- electro-mechanical malfunctions affecting equipment which lead to stoppage of the conveyor system, or faults in the conveyors themselves.
- Malfunctions making manual intervention necessary as an emergency strategy.

Organizational malfunctions
The following organizational malfunctions are taken into account as appropriate:

- Malfunctions due to missing parts.
- Malfunctions due to faulty components (the products and components to be processed demonstrably fail to meet the quality standards required by the Client (incorrect material, inadequate quality)).
- Non technically based electro-mechanical malfunctions in the conveyor technology, e.g. caused by passing through a photoelectric barrier in an enclosed area, leading to stoppage of the conveyor system and therefore of the line.
- Malfunctions in the use of handling or mechanical devices (e.g. flaps, risk of entrapment, Q-stop etc.), which can lead to line stoppage.
- Effects of human error or failure, e.g. incorrect equipment operation.
- Pressing the EMERGENCY OFF button

5.5. Supplier Evaluation

After completion of the project, Mercedes-Benz AG will carry out an evaluation of the project according to the following criteria:

- Tender quality
- Self-organisation of the supplier in projects
- Sub-supplier quality
- Compliance with technical standards (according to Enterprise-Content-Management Enterprise)
- Compliance with on-site requirements (SOS) and environmental protection guidelines

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- Number of deficiencies and/or duration of remedial work (more critical individual assessment is decisive)
- Quality and completeness of project documentation (especially if relevant to acceptance)
- Adherence to schedule
- Response to supplementary work requirements
- Achievement of specifications with respect to cycle time/technical availability and OK quality (more critical individual assessment is decisive)
- Innovativeness
- Compliance with maintenance and servicing intervals according to plan

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